COURSE SYLLABUS

ENGR 3303: Controls Engineering

Fall 2026

RANDOLPH COLLEGE

Instructor	Dr. Huan Phan, Assistant Professor of Robotics and Mechatronics Engineering
Email	hphan206@randolphcollege.edu
Class meetings	**Mon., Tue., Thu., Fri. • 10:55 a.m.–12:10 p.m. • Martin Science Building, Room 208**
Course dates	October 26–December 14, 2026
Credit hours	4
Prerequisites	PHYS 1115 and ENGR 1105
Office hours	By appointment; weekly times posted on the course website
Course site	`huantianh.github.io/courses/control-systems` Announcements, lecture notes, laboratory instructions, and schedule updates

Course Description

This course introduces the fundamentals of feedback control through the practical application of DC motor speed control. Students learn how a simple motor system responds, how feedback improves performance, and how a proportional or proportional-integral controller can be used to regulate motor speed.

The course provides an introduction to linear systems, transfer functions, Laplace transforms, stability, feedback, transient response, basic control design, discrete-time implementation considerations, and introductory frequency-domain concepts. Python is used to record experimental data, create plots, simulate simple motor models, and support the design and testing of the DC motor speed controller. Arduino code is used to implement the controller on the motor hardware.

Course Learning Outcomes

By the end of the course, a successful student will be able to:

1. explain the difference between open-loop and closed-loop control;
2. describe the basic input, output, reference, error, and feedback signals in a DC motor speed-control system;
3. use a simple transfer function to represent first-order motor behavior;
4. interpret a motor step response using rise time, settling time, and steady-state value;
5. use Python to record, visualize, and analyze open-loop and closed-loop motor behavior;
6. explain basic stability using system response and pole location;
7. explain how proportional and integral control affect motor-speed response;
8. implement and tune a simple P or PI motor speed controller;
9. explain basic sampling and practical controller implementation considerations; and

10. present and interpret experimental motor-control results using clear plots and engineering reasoning.

Course Materials

- Course notes and laboratory instructions provided by the instructor.
- A laptop with Python 3 installed.
- Free Python packages used in the course: NumPy, pandas, Matplotlib, SciPy, and pyserial.
- Arduino IDE for uploading motor-controller code.
- DC motor-control hardware provided by the instructor.

No textbook purchase is required unless announced by the instructor. Students should not purchase additional hardware or software for this course unless specifically instructed.

Course Organization and Expectations

The course meets four times per week for 75 minutes. The normal weekly structure is:

- **Monday – Lecture X.1:** Introduction of the week's main concept.
- **Tuesday – Lecture X.2:** Continued explanation, examples, and preparation for laboratory work.
- **Thursday – Lab X.1:** Guided Python analysis, implementation, or motor-testing activity.
- **Friday – Lab X.2:** Continue the same laboratory, interpret results, complete required work, and submit the lab.

Lab X.1 and Lab X.2 are two class meetings for one graded laboratory assignment. Week 5 is shortened because of Thanksgiving break and uses a condensed lecture-and-laboratory format. Weeks 1–6 contain the technical course material. Week 7 contains no new technical material and is reserved for final project preparation, presentations, demonstrations, peer evaluation, and course completion.

The DC motor speed-control system serves as the common application throughout the course. Each laboratory builds on previous work so that students gradually move from open-loop motor testing to a working feedback speed controller.

Assessment and Grading

Assessment category	Weight
Labs	50%
DC Motor Speed-Control Project and Presentation	30%
Quizzes and Concept Checks	10%
Teamwork and Peer Evaluation	10%
Total	**100%**

A	B	C	D	F
90–100%	80–89.9%	70–79.9%	60–69.9%	below 60%

Labs

Laboratory work is the primary method for applying course concepts. Thursday and Friday normally form one laboratory assignment.

The laboratory sequence includes:

1. Open-Loop DC Motor Speed Control
2. Motor Step Response and First-Order Model
3. Proportional Motor Speed Control
4. PI Motor Speed Control
5. Digital Controller Implementation and Filtering
6. Final Controller Validation and Project Preparation

Lab grades reflect preparation, completion of required tasks, correct use of Python, Arduino code, and hardware, interpretation of results, engineering reasoning, and clear documentation.

DC Motor Speed-Control Project and Presentation

The final project is the continuation of the laboratory work completed during Weeks 1–6. Students work in small groups to complete, demonstrate, and explain a functioning DC motor speed-control system.

Each group should be able to show:

- the DC motor and controller setup;
- open-loop motor behavior;
- the feedback-control structure;
- the P or PI controller used;
- target motor speed compared with measured motor speed;
- at least one clear performance plot;
- one problem encountered during development; and
- one improvement made to the final controller.

The project is evaluated on correct operation, appropriate use of feedback, quality of testing, interpretation of results, technical explanation, and the final presentation and demonstration.

Quizzes and Concept Checks

Short quizzes or concept checks may be given during lecture meetings. These assessments focus on the main ideas needed for laboratory work and the final project.

Topics may include:

- open-loop versus closed-loop control;
- basic transfer-function interpretation;
- first-order response;
- rise time, settling time, and steady-state error;
- basic stability;

- effects of P and PI control;
- sampling and implementation concepts; and
- basic interpretation of a frequency-response plot.

Teamwork and Peer Evaluation

Students will work in small groups during selected laboratory activities and the final DC motor speed-control project.

At the end of the course, students will confidentially evaluate their teammates based on preparation, participation, communication, reliability, respect for teammates, and contribution to the group's work.

Each student remains responsible for understanding the group's Python analysis, controller, hardware, test results, plots, and conclusions.

Laboratory Safety and Equipment Care

Follow all laboratory instructions and use tools and components only for their intended purpose. Wear safety glasses when instructed, secure long hair and loose clothing, disconnect power before changing wiring, and verify motor-driver and power connections before energizing hardware.

Motors and other high-current devices must not be powered directly from Arduino input/output pins. Use approved motor drivers and power supplies, maintain a common electrical reference when required, and return equipment in organized condition.

Course Policies

Attendance

Regular attendance is expected. In a seven-week course, one absence represents substantial material and laboratory time. Notify the instructor as early as possible and remain responsible for missed work.

Late and Make-Up Work

Unless stated otherwise, late work loses 10 percentage points per calendar day. Contact the instructor before the deadline when illness or an emergency prevents timely completion. Missed laboratory work may require an alternative assignment.

Teamwork and Individual Responsibility

Students may collaborate on approved laboratory and project activities, but each student must understand the group's Python analysis, controller implementation, test results, and conclusions.

Individual quizzes, concept checks, notebook entries, and reflections must represent the student's own work unless otherwise announced.

Communication and Conduct

Use Randolph College email and include ENGR 3303 in the subject line. Contribute to a respectful, inclusive, and productive learning environment. Phones and unrelated devices should be put away; laptops may be used for course work.

Randolph College Policies and Resources

Accommodations

Students seeking accommodations should contact Larvail Jones, Coordinator of Access Services, at ljones@randolphcollege.edu or 434-947-8132, then meet privately with the instructor after receiving an accommodation letter.

Title IX and Sexual Misconduct

Faculty members are mandatory reporters. Reports involving sexual misconduct or a campus crime must be shared with the Title IX Coordinator, Jaclyn Beard, at jbeard@randolphcollege.edu or 434-947-8031. The Counseling Center is a confidential resource at counselingcenter@randolphcollege.edu or 434-947-8130.

Honor Code

Students must submit their own work, identify collaborators, cite sources and software when required, and never invent calculations, simulation results, measurements, or test results. Violations will be handled under the Randolph College Honor Code. See `https://www.randolphcollege.edu/studenthandbook/`.

Artificial Intelligence

AI tools may be used for brainstorming, grammar, conceptual explanations, or debugging questions. They may not generate submitted calculations, Python-generated results, code, diagrams, laboratory reports, project documentation, quizzes, or reflections unless explicitly authorized.

Permitted use must be disclosed, and students remain responsible for accuracy, engineering judgment, safety, and originality.

Tentative Course Schedule

Weeks 1–6 introduce the technical material through the common application of DC motor speed control. The normal weekly structure is two lectures followed by two laboratory meetings. Week 5 is shortened because of Thanksgiving break. Week 7 contains no new technical material and is reserved for final project preparation, presentations, demonstrations, peer evaluation, and course completion.

Week	Date	Topic or activity
1	Mon., Oct. 26	**Lecture 1.1 – Introduction to Control Systems:** Control engineering, open-loop versus closed-loop control, feedback, desired motor speed, measured motor speed, error, and the DC motor as the common course system.
1	Tue., Oct. 27	**Lecture 1.2 – Basic Transfer Functions:** Brief Laplace-transform review, transfer-function concept, input and output signals, and the simple first-order form used to describe motor behavior.
1	Thu., Oct. 29	**Lab 1.1 – Open-Loop DC Motor Speed Control:** Connect the motor and motor driver, apply several PWM commands, measure motor speed, and record PWM-versus-RPM data.
1	Fri., Oct. 30	**Lab 1.2 – Open-Loop DC Motor Speed Control:** Complete the motor tests, create a PWM-versus-speed plot in Python, observe variation in motor speed, identify limitations of open-loop control, and submit Lab 1.
2	Mon., Nov. 2	**Lecture 2.1 – First-Order Motor Response:** Introduce the simplified model $G(s) = K/(\tau s + 1)$ and interpret gain K and time constant τ using motor-speed examples.
2	Tue., Nov. 3	**Lecture 2.2 – Motor Step Response and Basic Stability:** Step response, rise time, settling time, steady-state value, and stable versus unstable behavior using plots and simple examples.
2	Thu., Nov. 5	**Lab 2.1 – Motor Step Response and Data Logging:** Run a motor step test, record RPM directly from Arduino into Python, save the data as CSV, and observe motor speed versus time.
2	Fri., Nov. 6	**Lab 2.2 – First-Order Motor Model:** Estimate K and τ from measured data, simulate the first-order response in Python, compare model and experiment, discuss model limitations, and submit Lab 2.
3	Mon., Nov. 9	**Lecture 3.1 – Feedback and Proportional Control:** Feedback error, desired speed versus measured speed, proportional control, and the meaning of controller gain K_p.
3	Tue., Nov. 10	**Lecture 3.2 – Choosing Proportional Gain:** Effect of changing K_p, slow response, faster correction, steady-state error, and basic stability considerations.
3	Thu., Nov. 12	**Lab 3.1 – Proportional Motor Speed Control:** Implement closed-loop proportional control on Arduino, record reference, RPM, error, and PWM with Python, and test several values of K_p.
3	Fri., Nov. 13	**Lab 3.2 – Proportional Gain Tuning:** Compare several K_p tests in Python, evaluate steady-state error, response speed, and saturation, select a reasonable gain, and submit Lab 3.
4	Mon., Nov. 16	**Lecture 4.1 – Integral Action and PI Control:** Review the limitations of proportional control, introduce integral action and discrete PI implementation, and explain integral windup and basic anti-windup.
4	Tue., Nov. 17	**Lecture 4.2 – PI Tuning and Anti-Windup:** Explain the effect of K_i, identify poor PI tuning, use a safe tuning procedure, and apply integral clamping to reduce windup.

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Week	Date	Topic or activity
4	Thu., Nov. 19	**Lab 4.1 – PI Motor Speed Control:** Implement PI control on Arduino, record controller data in Python, compare P and PI behavior, and begin tuning K_i.
4	Fri., Nov. 20	**Lab 4.2 – PI Tuning and Comparison:** Test several K_i values, compare error, overshoot, saturation, and integral behavior in Python, select a PI controller, and submit Lab 4.
5	Mon., Nov. 23	**Lecture 5 – Digital Implementation and Filtering:** Explain sample time, discrete PI updates, encoder-count resolution, serial logging, filtered RPM measurements, and the tradeoff between smoothing and delay.
5	Tue., Nov. 24	**Lab 5 – Digital Implementation and Filtering:** Run the PI controller at a fixed sample time, compare raw and filtered RPM in Python, evaluate repeatability and practical limitations, and submit the shortened Lab 5.
5	Thu., Nov. 26	**No class – Thanksgiving break.**
5	Fri., Nov. 27	**No class – Thanksgiving break.**
6	Mon., Nov. 30	**Lecture 6.1 – Frequency Response for Motor Systems:** Introduce sinusoidal inputs, magnitude, phase, first-order corner frequency, and the basic purpose of Bode plots using Python-generated examples.
6	Tue., Dec. 1	**Lecture 6.2 – Final Controller Validation and Project Preparation:** Plan repeatable final tests, review troubleshooting, select required plots, document final controller settings, and prepare the Week 7 presentation and demo.
6	Thu., Dec. 3	**Lab 6.1 – Final Controller Validation:** Test the final controller at several safe reference speeds, record experimental data with Python, calculate performance metrics, and prepare comparison plots.
6	Fri., Dec. 4	**Lab 6.2 – Repeatability and Project Preparation:** Repeat the final controller test, optionally perform an instructor-approved disturbance test, finalize project figures, verify the live demo, and prepare results for Week 7.
7	Mon., Dec. 7	**Final Project Preparation:** Complete data analysis, finalize plots, organize project results, prepare presentation materials, and rehearse the motor demonstration.
7	Tue., Dec. 8	**Group Presentations:** Present the motor system, open-loop behavior, feedback structure, controller choice, target versus measured speed, final results, challenges, and improvements.
7	Thu., Dec. 10	**DC Motor Speed-Control Demonstrations:** Demonstrate the final feedback controller, target-speed tracking, measured motor speed, and stable operation.
7	Fri., Dec. 11	**Final Demonstrations and Peer Evaluation:** Complete remaining demonstrations, peer evaluation, discussion of group results, and project feedback.
7	Mon., Dec. 14	**Course Completion:** Final project submission, individual reflection, equipment return and cleanup, course wrap-up, and discussion of future control-system topics.

The instructor may revise this syllabus when necessary. Substantive changes will be announced in class and posted on the course website.